DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

DATA ANALYTICS

Submitted in partial fulfillment for the award of degree(*21INT82*)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Dharshan R Kumar (1DS21AI015)

Conducted at

CODTECH IT SOLUTIONS PVT.LTD

IT SERVICES & IT CONSULTING

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangelore-78

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CERTIFICATE

This is to certify that the Internship titled **"Data Analytics"** carried out by **Dharshan R Kumar [1DS21AI015]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

N.Santhosh

Signature of Reporting Head
Mr. Neela Santhosh Kumar
HR & Academic Head
CodeTech

Signature of Internship Coordinator
Assistant Prof. Kavya D N
Dept. of AIML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner	Signature with Date
1) ____________________	________________
2) ____________________	________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

DECLARATION

I, **Dharshan R Kumar,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **CODTECH IT SOLUTIONS PVT.LTD**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:12/04/2025 USN: 1DS21AI015

Place: Bangalore NAME:DharshanRKumar

OFFER LETTER

CODTECH IT SOLUTIONS PVT.LTD

IT SERVICES & IT CONSULTING

8-7-7/2, Plot NO.51, Opp: Naveena School, Hasthinapuram Central, Hyderabad , 500 079. Telangana

Internship Offer Letter

Date : 24/02/2025
Intern ID :CT04MEY

Dear Dharshan R Kumar

Congratulations on being selected for the **MACHINE LEARNING.** We at **CODTECH IT SOLUTIONS PVT.LTD** are thrilled to have you join our team. **This Online Internship will span 4 Months from February 27th, 2025 to June 27th, 2025.**

This internship is designed as an educational experience, focusing on learning, skill development, and gaining practical knowledge. As an intern, we expect you to:

1. Complete all assignments to the best of your ability.
2. Follow any lawful and reasonable instructions provided by your supervisors.
3. Participate actively in team meetings and discussions.
4. Provide regular updates on your progress.
5. Adhere to company policies and maintain a professional demeanor.
6. Collaborate effectively with team members and contribute to group projects.
7. Seek feedback and apply it to improve your performance.

We trust that you will approach all tasks with diligence and enthusiasm. We are confident that this internship will be an enriching experience for you. We look forward to working with you and supporting you in achieving your career aspirations

Best regards,

N. Santhosh

Neela Santhosh Kumar
Human Resources & Academic Head
CODTECH IT SOLUTIONS PRIVATE LIMITED

VERIFIED BY
CODTECH IT SOLUTIONS PVT LTD

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ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Neela Santhosh Kumar**, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Assistant **Prof. Kavya D N,** Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Dharshan R Kumar[1DS21AI015]

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CHAPTER 1
INTRODUCTION

This internship focuses on developing hands-on experience in key areas of data science, machine learning, and big data engineering through a practical, real-world project based on Amazon Prime-related data. The aim is to empower interns with skills that are highly sought after in the data and AI industry, while also helping them understand the end-to-end data workflow—from data ingestion to visualization and sentiment interpretation. The project is divided into four major tasks, each targeting a crucial area of modern data-driven applications.

This internship provides hands-on experience in data science, machine learning, and big data analytics using real-world Amazon Prime data. It is designed to build practical skills across the end-to-end data pipeline—from data processing to visualization and sentiment analysis. The project is divided into four core tasks, each focusing on a crucial domain of modern data applications.

Task 1: Big Data Analysis introduces interns to distributed computing with PySpark, teaching how to handle large datasets, clean data, and extract actionable insights. Interns learn to scale data operations efficiently—an essential skill in data engineering and enterprise analytics.

Task 2: Predictive Analysis focuses on building machine learning models. Interns perform data preprocessing, feature selection, and model evaluation to predict outcomes like content type or popularity. This task builds a solid foundation in model development and evaluation used in real-world AI applications.

Task 3: Dashboard Development trains interns in data visualization tools such as Tableau or Power BI. The focus is on converting complex data into interactive dashboards that support decision-making. It enhances skills in reporting and communicating insights to both technical and business audiences.

Task 4: Sentiment Analysis equips interns with NLP techniques to classify user feedback from reviews or comments. This includes text cleaning, feature extraction, and training sentiment classifiers to interpret customer emotions—vital for marketing and brand analysis.

Together, these tasks prepare interns for real-world roles in data analytics, AI, and business intelligence, combining technical proficiency with problem-solving and storytelling through data.

CHAPTER 2
DESCRIPTION OF ORGANIZATION

CODTECH IT SOLUTIONS PVT.LTD

CODTECH IT SOLUTIONS PRIVATE LIMITED is a dynamic IT services and consulting firm based in Hyderabad, established in 2023. The company specializes in offering a wide range of digital solutions such as web and mobile application development, ui/ux design with maintenance, digital marketing strategies, and customized IT training programs. CODTECH's core mission is to empower startups and small businesses by delivering accessible, scalable, and modern technological solutions that foster digital transformation and professional growth.

In the domain of Data Analytics, CODTECH stands out for its robust service offerings and forward-thinking approach. The company provides cloud analytics modernization, enabling clients to future-proof their tech infrastructure by adopting scalable, cloud-native technologies. These services are designed to streamline operations, automate reporting, and optimize decision-making processes. By transforming raw, unstructured datasets into valuable business insights, CODTECH helps organizations unlock growth opportunities, improve efficiency, and enhance customer engagement.

The company's data analytics services encompass:

- Data Engineering and ETL pipeline design
- Business Intelligence (BI) reporting and dashboards
- Cloud-based Analytics using platforms like AWS, Azure, and GCP
- Data Visualization with tools such as Tableau and Power BI

Interns at CODTECH gain valuable exposure by working on live datasets—such as the Amazon Prime Video dataset—and contribute to full-cycle analytics projects involving:

- Big Data processing using PySpark or Dask
- Predictive modeling using machine learning algorithms
- Interactive dashboard development
- Sentiment analysis using NLP techniques

CHAPTER 3
TASKS

Task 1: Big Data Analysis Using PySpark

Goal: Perform scalable data processing and analysis on large datasets (like Amazon Prime Video data) using PySpark to extract business insights**.**

Big Data Analysis involves examining large and complex datasets that cannot be processed efficiently using traditional data handling tools like Excel, pandas, or SQL alone. As data volume, velocity, and variety increase, tools like Apache Spark become essential.

PySpark is the Python interface to Apache Spark — a powerful open-source distributed computing engine. It allows developers and data analysts to run distributed data processing jobs over clusters, making it ideal for analyzing large volumes of data quickly and efficiently.

1. Load the Dataset

- Use SparkSession to initiate the environment.
- Load the CSV file using spark.read.csv() with header and schema inference.
- Automatically partitions data across multiple executors for parallel processing.
- Supports loading data from cloud sources (e.g., AWS S3, Google Cloud Storage) for enterprise-scale deployment.

2. Data Cleaning

- Cleaning is essential to ensure data quality
- Null Handling: Drop or impute rows/columns with missing values in important fields like title, genre, type.
- Date Conversion: Convert string dates (e.g., date_added) into Spark's DateType for date-based queries.
- Deduplication: Remove duplicate rows using .dropDuplicates() to ensure accuracy.

3. Data Processing and Aggregation

Using Spark's transformation operations like groupBy(), agg(), and filter():

- Group by Genre: Count how many movies/shows are available per genre.
- Country-wise Count: Analyze how content is distributed across different countries.
- Year-wise Trend: Count how many titles were added per year using year(date_added) function

4. Data Aggregation and Insight Extraction

Using Spark SQL functions, we can extract business intelligence from the dataset:

- Count the number of titles in each genre using the listed_in column to identify what content is most in demand.
- Grouping by year(date_added) shows how content availability has grown over time.
- By parsing the duration field, we calculate the average length of movies and the average number of seasons for TV shows.

Code:

```
from pyspark.sql import SparkSession
spark = SparkSession.builder \
    .appName("Amazon Prime Big Data Analysis") \
    .getOrCreate()
# Load the dataset
df = spark.read.option("header", "true").csv("amazon_prime_titles.csv")
# Preview data
df.show(5)
from pyspark.sql.functions import col, to_date
# Drop nulls in essential columns
df_clean = df.dropna(subset=["title", "date_added", "listed_in", "duration", "type"])
# Convert date_added to DateType
df_clean = df_clean.withColumn("date_added", to_date(col("date_added"), "MMMM d, yyyy"))
# Drop duplicates
df_clean = df_clean.dropDuplicates()
from pyspark.sql.functions import split, explode
# Split 'listed_in' into individual genres
df_genres = df_clean.withColumn("genre", explode(split(col("listed_in"), ", ")))
# Count and sort top genres
top_genres = df_genres.groupBy("genre").count().orderBy(col("count").desc()).limit(10)
top_genres.show()
```

Conclusion

This task showcases the efficiency of PySpark in handling and analyzing large-scale datasets. It enables actionable insights like genre trends and yearly content growth, aiding strategic decisions for content platforms. Compared to pandas, PySpark offers superior scalability, speed, and fault tolerance, making it ideal for big data analytics.

Task 2: Predictive Analysis Using Machine Learning

Goal: Build a machine learning model to predict outcomes (e.g., movie/show popularity or success) using classification or regression techniques. This involves feature engineering, model training, evaluation, and deriving insights.

Predictive analysis is a branch of data analytics that uses historical data to make future predictions. In this task, the aim is to train a machine learning model on a cleaned dataset (e.g., Amazon Prime Video metadata) to classify content (like movies/shows) based on specific features such as duration, country, genre, etc.

We use Python's **scikit-learn** library due to its simplicity, robustness, and wide adoption in the data science community. It offers tools for data preprocessing, model selection, evaluation, and deployment.

1. Dataset Preparation

The first step in any machine learning task is to gather and prepare the dataset in a structured form. For this task, we are using the cleaned dataset from Task 1, which consists of metadata about Amazon Prime Video titles. The focus is to predict whether a content title is a Movie (1) or TV Show (0) based on several features.

Key Steps:

- Load Dataset: The dataset is loaded using pandas.read_csv() for preprocessing before model training.
- Select Features: From the available columns, we choose:
- type: Target variable to classify (Movie vs. TV Show).
- duration: Important for distinguishing between movies (in minutes) and TV shows (in seasons).
- release_year: Useful to capture content trend changes over years.
- country: Often determines the type of content produced (regional variations).
- listed_in: Represents genre and can indicate the type of title.
- Create Label Column: Convert the type column into a binary label: Movie = 1, TV Show = 0.

2. Data Preprocessing

Preprocessing is essential to convert raw data into a machine-readable format and to prepare features that are suitable for ML algorithms.

Steps Included:

- One-Hot Encoding:
  - Categorical columns like country and listed_in are encoded into binary vectors.
  - Helps ML models understand non-numeric data.
- Duration Parsing:
  - The duration field varies:
    - Movies are written like “90 min” → extract 90.
    - Shows are written like “3 Seasons” → can be converted to estimated minutes.

- We create a new numerical column duration_mins.

- Null Handling:
  - Drop rows with missing data in critical fields like duration, type, country, or listed_in.
- Scaling:
  - Apply StandardScaler to numerical columns like duration_mins and release_year to bring them into a standard range for better performance of gradient-based models like Logistic Regression.

3. Model Selection and Training

Once the data is clean and structured, the next phase is to choose a suitable algorithm and train the model.

Algorithm: Logistic Regression:

- Ideal for binary classification tasks.
- Interpretable and fast to train, making it suitable for a beginner to intermediate use case.

Model Workflow:

- Train-Test Split: Split the data (e.g., 80% train, 20% test) to evaluate how well the model generalizes to unseen data.
- Model Training: Fit the logistic regression model to the training data.The model learns the relationship between the features and the binary label.
- Evaluation Metrics: Shows how well the model distinguishes between movies and shows.
- Gives precision, recall, and F1-score, helping assess performance in detail.

4. Code Implementation

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransforrm

from sklearn.pipeline import Pipeline

# Load dataset
df = pd.read_csv("amazon_prime_titles.csv")
# Filter necessary columns
df = df[['type', 'duration', 'country', 'release_year', 'listed_in']]
df.dropna(inplace=True)
# Feature engineering
```

```python
df['duration_mins'] = df['duration'].apply(lambda x: int(x.split()[0]) if 'min' in x else 60)
df['label'] = df['type'].apply(lambda x: 1 if x == 'Movie' else 0)
# Define features and label
X = df[['country', 'release_year', 'listed_in', 'duration_mins']]
y = df['label']
# Preprocessing pipeline
categorical = ['country', 'listed_in']
numerical = ['release_year', 'duration_mins']
preprocessor = ColumnTransformer(transformers=[
    ('cat', OneHotEncoder(handle_unknown='ignore'), categorical),
    ('num', StandardScaler(), numerical)])
# Define pipeline
pipeline = Pipeline(steps=[ ('preprocess', preprocessor),  ('classifier', LogisticRegression())])
# Split data and train
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
pipeline.fit(X_train, y_train)
# Predictions and Evaluation
y_pred = pipeline.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Task 3: Dashboard Development Using Tableau

Goal: To design and build an interactive, user-friendly dashboard using real-world data (Amazon Prime Video dataset) to deliver actionable business insights. This task focuses on the visualization of previously derived data insights—like genre distribution, yearly trends, and content duration—into an accessible and insightful format for decision-makers.

1. Dataset Preparation

Before creating visualizations, the dataset must be cleaned and structured properly.

Key Preprocessing Tasks:

- Data Cleaning: As performed in Task 1, we remove rows with missing values in critical columns like title, listed_in, date_added, duration, and type.
- Date Parsing: Convert the date_added column to proper DateType to allow time-based filtering in the dashboard.
- Genre Extraction: Extract individual genres from the listed_in column for genre-wise visualizations.
- Duration Transformation: Parse the duration column to extract:
    - Movie durations in minutes.
    - Number of seasons for TV shows.

Software Used:

- Tableau / Power BI: For drag-and-drop dashboard development with built-in filtering, slicers, and publishing options.
- Dash (Optional for coders): Python framework for fully customizable dashboards.

2. Key Visualizations

These visualizations convert raw insights into powerful visuals:

1. Top 10 Most Popular Genres

- Chart Type: Bar Chart
- Logic: Count titles under each genre.
- Insight: Shows user content preferences (e.g., Drama, Comedy, Documentary).

2. Content Added Per Year

- Chart Type: Line Chart
- Logic: Group data using year(date_added).
- Insight: Understand platform content growth trends.

3. Country-wise Content Distribution

- Chart Type: Map or Horizontal Bar Chart
- Logic: Count titles from different production countries.
- Insight: Understand regional content availability.

4. Average Duration Analysis

- Chart Type: Box Plot or Histogram
- Logic: Analyze average movie length and number of seasons for TV shows.

5. Insight: Determines content length trends and user viewing time expectations.
   Content Type Comparison

- Chart Type: Pie Chart
- Logic: Proportion of Movies vs. TV Shows.
- Insight: Understand platform focus on content types.

3. Filters and Interactivity

To enhance user experience, the dashboard includes:

- Date Slider: Filter content based on the year added.
- Genre Filter: Drill down into specific genres.
- Country Selector: View content based on country of origin.
- Content Type Toggle: Switch between TV shows and Movies.

These features help business users or content managers to explore data in multiple dimensions and derive **custom insights**.

4. Deliverables

Fully Functional Dashboard File

A complete, interactive dashboard created using Tableau (.twbx), Power BI (.pbix), or Dash (Python script). The dashboard included multiple sheets/pages covering various aspects of Amazon Prime data (e.g., top genres, year-wise trend, content by country, and duration insights).

Each visualization was connected to the underlying data source, allowing for interactivity (e.g., filters, slicers, and drill-downs).

Dashboard followed a clean and user-friendly layout, making it accessible to both technical and non-technical stakeholders.

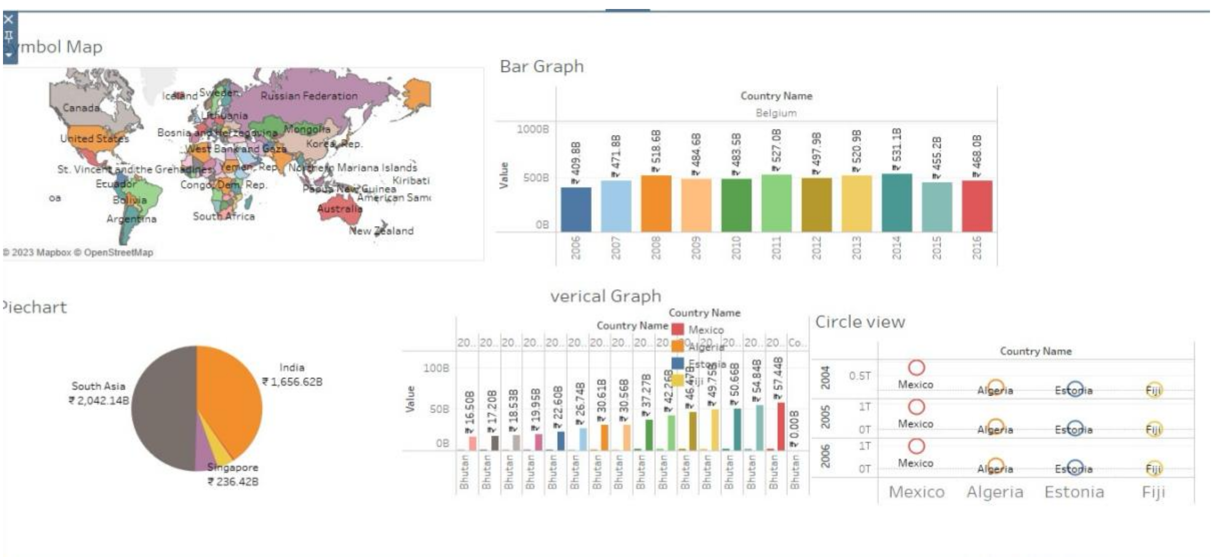
Screenshots of All Visualizations with Explanations

High-resolution screenshots of every chart/graph were included as documentation.

Each visualization was accompanied by a brief explanation describing:

- What the chart is displaying
- Why it is important

Fig-3.1: Amazon prime video dashboard



Conclusion

This task showcases the value of visual storytelling in data analytics. The dashboard turns complex, raw data into clear, actionable insights that decision-makers can use to identify genre trends, plan content acquisition, and target regional audiences.

Interactive dashboards like these bridge the gap between data and action, empowering both technical and non-technical users with real-time insights for strategic decision-making

Task 4: Sentiment Analysis Using Natural Language Processing

Goal: To analyze textual data (such as reviews or tweets) and classify them as positive, neutral, or negative using Natural Language Processing techniques. This task aims to derive public sentiment about content on platforms like Amazon Prime, helping businesses make data-driven decisions based on user feedback.

1. Dataset Preparation

The dataset used for this task includes user reviews or comments on Amazon Prime shows or movies. Each entry consists of the title of the content, the user's review text, and optionally a numeric rating provided by the user, usually on a scale of 1 to 5.

The dataset helps us analyze how viewers feel about specific titles — whether they loved them, were neutral, or disliked them. This kind of sentiment analysis is especially useful for platforms like Amazon Prime to determine which shows are well-received and which ones may need improvements or better marketing.

Before diving into any Natural Language Processing, we ensure that:

- The data format is consistent
- The review text is present and meaningful (non-empty and non-null)
- The ratings are usable to generate labels for supervised learning

If no numeric ratings are available, we could also use external sentiment scoring APIs or rule-based systems to label sentiment. However, in this case, since we assume ratings are provided, we use them to label each review as positive, neutral, or negative.

2. Data Preprocessing

Preprocessing textual data is one of the most critical steps in any NLP task. Raw text contains a lot of unnecessary characters, inconsistencies, and variations that can confuse a machine learning model. Here's what we do step-by-step:

- Convert all text to lowercase: This helps avoid duplication of words that differ only in case (e.g., "Amazing" vs. "amazing").
- Remove punctuation and special characters: These symbols don't carry sentiment value and can

distort token analysis.

- Tokenization: This is the process of splitting a sentence into individual words or tokens. This is fundamental for further transformations like stemming or lemmatization.
- Stopword removal: Words like "the", "is", "a", etc., are very common but don't contribute to the overall sentiment. Removing them reduces noise.
- Lemmatization: Instead of reducing a word to its root (like stemming), lemmatization converts a word to its base dictionary form (e.g., "running" → "run", "better" → "good"), which is more accurate for sentiment detection.

Finally, based on the ratings, we assign sentiment labels:

- Positive: Reviews with a rating of 4 or 5
- Neutral: Reviews with a rating of 3
- Negative: Reviews with a rating of 1 or 2

These labels are crucial for training the classification model in the next step.

Code Implementation:

```
import pandas as pd
import numpy as np
import re
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

# Download necessary NLTK data
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('wordnet')

# Load the dataset
df = pd.read_csv("amazon_prime_reviews.csv")

# Label sentiment
def label_sentiment(rating):
    if rating >= 4:
        return 'positive'
    elif rating == 3:
        return 'neutral'
    else:
        return 'negative'

df['sentiment'] = df['rating'].apply(label_sentiment)

# Preprocessing function
lemmatizer = WordNetLemmatizer()
```

```
stop_words = set(stopwords.words('english'))


def preprocess(text):
    text = text.lower()
    text = re.sub(r"[^a-zA-Z]", " ", text)
    words = nltk.word_tokenize(text)
    words = [lemmatizer.lemmatize(w) for w in words if w not in stop_words]
    return " ".join(words)

df['cleaned_review'] = df['review_text'].apply(preprocess)
df[['review_text', 'cleaned_review', 'sentiment']].head()
```

3. Model Selection and Training

After preprocessing the data, the cleaned text is still not machine-readable in a numerical form. Therefore, we apply TF-IDF Vectorization.

TF-IDF (Term Frequency–Inverse Document Frequency) gives weight to important words by considering how frequently they appear in a specific review compared to how common they are across the entire corpus. This results in a more meaningful representation of the review content, emphasizing words that are unique and potentially sentiment-bearing.

Next, we split the dataset into training and testing sets, commonly using an 80:20 ratio. This means 80% of the data is used to train the model, and 20% is held back to evaluate its performance.

For this task, we use Logistic Regression, which is a simple and effective classification algorithm especially useful for binary or multi-class text classification problems. It calculates probabilities and assigns classes based on a logistic function, making it interpretable and fast.

After the model is trained, we evaluate it using metrics like:

- Accuracy – How many predictions were correct overall
- Confusion Matrix – Shows where the model predicted incorrectly
- Precision, Recall, F1-score – Gives more granular insight into how well the model distinguishes between classes

All of these evaluation steps help us understand if the model is ready for production or if further tuning is needed.

```
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix

# Vectorization
tfidf = TfidfVectorizer(max_features=3000)
X = tfidf.fit_transform(df['cleaned_review'])
```

```
y = df['sentiment']
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Model training
model = LogisticRegression()
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Evaluation
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

4. Insight Extraction and Visualization

Sentiment Distribution:

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.countplot(x='sentiment', data=df)
plt.title("Sentiment Distribution in Amazon Prime Reviews")
plt.show()
```

Word Clouds per Sentiment:

```
from wordcloud import WordCloud
for sentiment in ['positive', 'neutral', 'negative']:
    text = " ".join(df[df['sentiment'] == sentiment]['cleaned_review'])
    wc = WordCloud(width=800, height=400, background_color='white').generate(text)
    plt.figure(figsize=(8,4))
    plt.imshow(wc, interpolation="bilinear")
    plt.axis('off')
    plt.title(f"Word Cloud for {sentiment} Reviews")
    plt.show()
```

5. Deliverables

The final output for this task is a comprehensive Python Notebook that includes the following components:

Data Preprocessing

- Cleaned and normalized the textual review data
- Applied text transformation techniques: lowercasing, punctuation removal, stopword filtering, tokenization, and lemmatization

- Generated sentiment labels (positive, neutral, negative) based on user ratings

Model Building

- Implemented TF-IDF vectorization to convert cleaned text into numerical format
- Trained a Logistic Regression model to perform multi-class sentiment classification
- Used an 80/20 train-test split to validate model performance

Evaluation and Metrics

- Calculated key performance metrics:

1. Accuracy Score to assess overall correctness
2. Confusion Matrix to visualize class-wise prediction distribution
3. Precision, Recall, F1-Score to evaluate the model's sensitivity to each class Visualizations

- Bar Charts displaying sentiment distribution across the dataset

Conclusion

This sentiment analysis task highlights the ability of Natural Language Processing (NLP) and machine learning to extract deep emotional intelligence from unstructured user feedback. By classifying viewer reviews into positive, neutral, or negative categories, this analysis enables content producers, marketers, and platform administrators to:

- Understand audience sentiment at scale
- Measure satisfaction for specific titles
- Identify trends or recurring criticisms

Using a scalable approach powered by TF-IDF and Logistic Regression, the model can handle large volumes of text with high accuracy. The automation of sentiment detection not only reduces manual effort but also provides timely and actionable insights to enhance content strategy and user engagement on streaming platforms like Amazon Prime.

Overall Summary of Internship Tasks

This internship offered a holistic learning experience in the domain of Data Analytics, covering essential pillars of modern data-driven decision-making. Beginning with Task 1: Big Data Analysis using PySpark, we explored scalable data processing techniques using distributed computing. PySpark enabled efficient handling of large datasets, extracting meaningful patterns such as popular genres, content growth trends, and duration analysis from Amazon Prime data. Moving to Task 2: Predictive Analysis using Machine Learning, we built a classification model to predict whether a title is a movie or TV show based on multiple features. This task enhanced our understanding of data preprocessing, feature selection, and model

evaluation using supervised learning techniques. In Task 3: Dashboard Development, we designed an interactive, user-friendly dashboard using Power BI to visualize key metrics such as genre popularity, year-wise content additions, and country-specific distributions. This visual representation transformed raw data into clear, actionable insights for stakeholders. Finally, in Task 4: Sentiment Analysis using NLP, we applied natural language processing to viewer reviews to classify sentiments as positive, neutral, or negative. The use of TF-IDF, Logistic Regression, and visual tools like word clouds provided a powerful method to understand user feedback and emotional responses at scale.

Collectively, these tasks demonstrate the interconnected roles of big data processing, machine learning, data visualization, and natural language understanding in the world of data analytics. This multi-faceted exposure helped bridge the gap between theoretical knowledge and practical implementation, preparing me to solve real-world business problems through data.

CHAPTER 4
SCOPE OF WORK

This internship project is designed to offer in-depth, hands-on exposure to **a** wide range of data science and big data technologies through the lens of a real-world application—analyzing and interpreting data from Amazon Prime Video. The scope of this project extends across the complete data pipeline, starting from raw data ingestion and cleaning to advanced machine learning and visualization. The work is divided into four key tasks, each aligned with the industry-standard practices of data engineering, data science, business intelligence, and natural language processing.

1. Big Data Analysis Using PySpark

This phase covers the large-scale data processing and exploratory data analysis (EDA) using Apache Spark's Python API—PySpark. The scope includes:
Data Ingestion: Reading the dataset in distributed mode using SparkSession and loading large CSV files with inferred schema.
Data Cleaning: Handling missing/null values, transforming string-based dates to date objects, removing duplicates, and structuring fields for further analysis.
Genre Analysis: Splitting multi-valued genre strings into individual rows and performing frequency analysis to find the top 10 most common genres across titles.
Year-Wise Trends: Extracting the year from date_added to analyze content release and addition trends over the years.
Duration Parsing: Differentiating between movies and TV shows and calculating average movie lengths and average number of seasons in shows.
Scalability Demonstration: Emphasizing how PySpark can process millions of records efficiently across clusters, highlighting the difference between PySpark and traditional pandas.
This component teaches how to deal with data at scale, a crucial skill in industries like OTT platforms, retail analytics, and digital media services.

2. Predictive Analysis Using Machine Learning

In this stage, the focus is on designing a supervised machine learning pipeline to predict content types (Movie or TV Show) based on available metadata.

Data Preparation:

- Filtering relevant features such as type, duration, release_year, listed_in, and country.
- Creating binary classification labels (Movie → 0, TV Show → 1).
- Parsing duration text (e.g., "90 min", "2 Seasons") into numeric form.
- Handling categorical features using one-hot encoding (genre, country).

Model Building:

- Using Logistic Regression for binary classification.
- Splitting the dataset into training and test sets (80/20).
- Transforming textual and categorical features using TF-IDF and StandardScaler.
- Training the model using scikit-learn and tuning for better performance.

Model Evaluation:

- Accuracy, precision, recall, F1-score.
- Confusion matrix to understand class-wise performance.
- Cross-validation to ensure generalization.
- This task teaches predictive modeling, model evaluation, feature engineering, and the logic of classification systems commonly used in recommendation engines and content categorization.

3. Dashboard Development Using Power BI or Tableau

This segment focuses on translating raw data and machine-generated outputs into meaningful business visuals that can help stakeholders make informed decisions.

KPI Identification:

- Total number of titles.
- Top 10 genres by frequency.
- Country-wise distribution of content.
- Year-wise content addition growth.
- Dashboard Features:
- Filters for Type, Country, Genre, Release Year.
- Interactive bar graphs, pie charts, line graphs.
- Trend visualization using time-series data.
- Genre distribution heatmaps.

Tool Use:

- Either Power BI or Tableau is used to connect to the cleaned dataset (CSV or database).
- Dynamic dashboards are built to allow viewers to interact with and filter content.
- This task enhances the intern's ability to build user-friendly interfaces, transform data into business intelligence, and use visual storytelling to present insights to decision-makers.

4. Sentiment Analysis Using NLP

This part involves analyzing viewer sentiments from user reviews, providing qualitative insights into how users perceive Amazon Prime content.
Dataset Used: A CSV file containing user reviews and ratings for various Amazon Prime titles.
Text Preprocessing:
Converting text to lowercase.
Removing punctuation, stopwords, special characters.
Tokenization and lemmatization using NLTK or spaCy.
Labeling Sentiments:

- Rating ≥ 4 → Positive
- Rating = 3 → Neutral
- Rating ≤ 2 → Negative

Model Implementation:

- Training a Logistic Regression classifier on sentiment labels.
- Splitting the data into training and testing sets.
- Evaluating the classifier using accuracy, precision, and F1-score.

Visualization:

- Word clouds for positive and negative sentiments.
- Bar charts showing sentiment distribution per genre or title.
- This task provides hands-on experience with natural language processing, text classification, and opinion mining, crucial for businesses that rely on customer feedback.

5. Full Data Pipeline Integration

All individual tasks contribute to an integrated data pipeline:
Raw data from CSV → Preprocessed using PySpark → Analyzed via ML and NLP → Visualized using Dashboards.
This end-to-end workflow represents real-world data science projects in industry settings.
Interns learn to build modular, reusable components across analytics, modeling, and visualization.

6. Skill Enhancement and Industry Readiness

This internship doesn't only focus on tools—it's structured to build real-time decision-making skills, logical problem-solving, and the ability to work on multi-disciplinary projects. Interns get experience with:

- Python, PySpark, Scikit-learn, NLTK
- Power BI / Tableau
- Real-world project planning
- Communicating insights to technical and non-technical audiences
- Handling structured and unstructured data

7. Collaboration, Documentation, and Version Control

This internship also emphasized the importance of working in a collaborative and reproducible manner—mimicking professional data science team environments.

- Interns documented each stage of the workflow in Jupyter Notebooks or Markdown reports, explaining code, logic, and outcomes.
- Git and GitHub were optionally introduced for version control, allowing tracking of changes, backup, and collaborative code reviews.
- Clear documentation enabled smooth handover of work, facilitated learning for peers, and enhanced code readability and project transparency.
- This approach nurtured best practices followed in real-world teams, like reproducibility, modularity, and iterative improvements.

8. Deployment and Future Scalability Considerations

While the internship focused on analysis and modeling, it also highlighted how such projects can evolve into scalable, production-level systems.

- Interns explored how machine learning models could be saved using joblib or pickle for future use or integration into web applications or dashboards.
- The importance of APIs and cloud deployment (like AWS or Streamlit) was introduced as an optional extension to expose sentiment analysis or predictions to users.
- Scalable dashboard designs were discussed, especially how dashboards could be refreshed dynamically using data from databases or live sources.
- This phase built awareness of deployment pipelines, containerization, and how models and dashboards can be integrated into business tools.

CHAPTER -5

LEARNING OBJECTIVES

This internship is structured to provide a real-world, end-to-end experience in the fields of Big Data Analytics, Machine Learning, Dashboard Development, and Natural Language Processing (NLP). Using Amazon Prime-related datasets, interns gain exposure to handling diverse data types and technologies, helping them become well-rounded data professionals. The following are the detailed learning objectives:

1. Understanding Big Data Concepts and Tools

- Learn the fundamentals of Big Data, including its characteristics: volume, velocity, and variety.
- Gain hands-on experience with PySpark, a powerful tool for distributed data processing in real-time.
- Understand how to load, clean, and transform large datasets that exceed in-memory capacity using Spark's DataFrame API.
- Learn about partitioning, lazy evaluation, and fault tolerance in distributed systems.
- Develop skills to extract aggregated insights and trends from large datasets efficiently.

2. Developing Predictive Models Using Machine Learning

- Master the full machine learning pipeline, from raw data to predictions.
- Learn how to perform data preprocessing, handle categorical/numerical features, and apply encoding and scaling techniques.
- Understand how to build models such as Logistic Regression or Random Forest for classification tasks.
- Learn to split datasets using train-test validation, avoid overfitting, and optimize models using hyperparameter tuning.
- Evaluate models using key metrics: accuracy, confusion matrix, precision, recall, F1-score, and ROC-AUC

3. Creating Interactive Dashboards for Data Storytelling

- Gain expertise in designing dashboards using Tableau, Power BI, or Dash.
- Learn to transform raw numbers into meaningful visual stories that support business decisions.
- Create visuals such as bar charts, pie charts, heatmaps, line graphs, and filters.

- Learn about data blending, calculated fields, interactivity (like slicers, drill-downs), and layout best practices.
- Understand the importance of real-time updates and how dashboards are used in monitoring KPIs and performance metrics.

4. Applying Natural Language Processing for Sentiment Analysis

- Understand the basics of text mining and natural language processing using Python libraries like NLTK or spaCy.
- Learn how to clean and preprocess unstructured text: lowercasing, removing stopwords, punctuation, lemmatization, etc.
- Implement TF-IDF vectorization and train machine learning models to classify user reviews into positive, neutral, or negative sentiments.
- Extract user opinions to gain insights into viewer satisfaction, content popularity, or improvement areas.
- Visualize the results using word clouds, bar plots, and confusion matrices.

5. Enhancing Critical Thinking and Problem-Solving Abilities

- Apply data-driven reasoning to formulate questions, test hypotheses, and derive actionable solutions.
- Strengthen analytical skills by learning how to identify patterns, correlations, and anomalies in large datasets.
- Learn how to combine technical skills with business acumen to solve real-world industry problems.
- Build confidence in using data tools and techniques to support data storytelling, reporting, and strategic planning.

6. Mastering End-to-End Data Project Lifecycle

- Understand how to initiate, plan, and execute a complete data science project.
- Learn about project documentation, code structuring, pipeline design, and reporting.
- Experience collaboration in a team-like environment—dividing tasks between data preparation, model development, and visualization.
- Gain insights into real-world data roles and responsibilities across data engineers, analysts,and scientists.
- Develop presentation skills to effectively communicate findings to technical and non-technical stakeholders.

7. Enhance Communication and Data Storytelling Skills

Develop the ability to interpret and convey complex data-driven findings in a simplified and visually appealing manner. This includes learning how to choose the right type of visual (bar charts, heatmaps, word clouds, etc.), write executive summaries, and tailor messages to both technical and non-technical audiences. Effective communication is critical for making data insights actionable. Interns will also practice presenting their dashboards and reports in a way that drives decision-making and aligns with business goals.

8. Understand End-to-End Project Workflow in Data Science

Acquire hands-on experience with the complete life cycle of a data science project. This includes data acquisition (from raw sources like CSV files or APIs), preprocessing and cleaning, exploratory data analysis, model selection and evaluation, visualization, and report generation. Interns will learn how each stage feeds into the next, emphasizing the importance of maintaining consistency, data integrity, and clarity throughout the process. This comprehensive workflow exposure is essential for preparing candidates for real-world roles in data analytics, machine learning, or full-stack data science development.

CHAPTER 6
CHALLENGES AND SOULTION

During the internship, several practical challenges were encountered while working with Amazon Prime data across different tasks. These challenges tested both technical and analytical skills and provided deep learning opportunities. Below are six major challenges faced during the project and the solutions implemented to overcome them.

1. Handling Large Volumes of Data with PySpark

Challenge:

The Amazon Prime dataset used in Task 1 was too large to be processed using traditional tools like pandas or Excel. Loading such data into memory caused system lags and frequent crashes.

Solution:

The challenge was resolved by switching to PySpark, a distributed processing framework that can handle large datasets efficiently. Spark's lazy evaluation, resilient distributed datasets (RDDs), and parallel computing allowed smooth data processing. Operations such as filtering, grouping, and aggregation were distributed across multiple cores, drastically improving performance.

2. Complex Data Cleaning and Inconsistent Formatting

Challenge:

The dataset had inconsistent date formats, missing values, and ambiguous entries in columns like duration, date_added, and country. This made transformation and analysis difficult.

Solution:

A combination of Spark SQL functions and custom logic was used to clean and format the data. Missing values in key fields were dropped, and custom parsing was applied to split text fields such as listed_in into genres. Date fields were converted into DateType using PySpark's to_date() function for easy filtering and trend analysis.

3. Feature Engineering for Machine Learning

Challenge:

In Task 2, transforming raw data into suitable features for predictive modeling was a challenge. Columns like duration had mixed values (e.g., "90 min" for movies and "3 Seasons" for TV shows), and listed_in had multiple genres per title.

Solution:
Custom feature extraction logic was implemented to standardize duration fields—converting "min" into integers and "seasons" into season counts. For genres, one-hot encoding was applied after splitting multi-genre entries using explode() and split(). This helped build a clean and structured dataset for machine learning models like Logistic Regression.

4. Difficulty in Interpreting Sentiments from Unstructured Reviews

Challenge:
In Task 4, performing sentiment analysis on free-form text reviews was challenging because the text was noisy, subjective, and often lacked consistent structure. Ratings didn't always align with sentiment expressed in the review text.
Solution:
Text preprocessing was performed thoroughly using NLTK—including converting text to lowercase, removing punctuation and stopwords, and applying lemmatization. A rule-based label assignment was used based on numeric ratings (positive, neutral, negative), and a TF-IDF vectorizer was applied to transform text for model training. The final classifier showed good performance in predicting sentiment accurately.

5. Visualizing Data Meaningfully in Dashboards

Challenge:
In Task 3, creating impactful visualizations from raw data was difficult, especially when trying to represent multiple KPIs and filter options in a limited screen space. Representing genre trends, year-wise additions, and average durations required careful design.
Solution:
Dashboards were designed using Power BI, with interactive slicers, dropdowns, and filters. Visuals were separated by categories (e.g., "Content by Genre," "Titles per Year," "Average Duration"), making them easier to interpret. DAX functions were used for calculated metrics, and visuals were styled with color-coding and tooltips to improve interactivity and user experience.

6. Ensuring Smooth Integration of All Tasks into One Workflow

Challenge:
Each task (Big Data Analysis, ML, Dashboarding, Sentiment Analysis) used different tools, technologies, and formats. Integrating all of them into a unified presentation and project flow was time-consuming and complex.

Solution:

The solution was to document and modularize each task, saving outputs in reusable formats such as .csv or parquet. Intermediate results were used as input for the next stage—for example, cleaned data from PySpark was exported and used in ML and visualization tasks. A well-structured folder hierarchy, naming conventions, and final documentation helped ensure a seamless integration and presentation of all components.

7. Choosing the Right Visualization Tool for Business Dashboards

Challenge:

With multiple dashboard tools available (Power BI, Tableau, Dash), selecting the most suitable one based on project goals, performance, and interactivity was initially confusing.

Solution:

A comparative analysis was done to understand the strengths of each tool. Power BI was chosen for its ease of use and integration with Microsoft services, while Tableau was used for deeper analytics with better design flexibility. Dash (Python-based) was also experimented with to allow full customization using code. Interns gained exposure to each platform, improving tool adaptability and decision-making skills.

8. Interpreting Sentiment Analysis Results with Accuracy

Challenge:

Assigning sentiment labels (positive, negative, neutral) from user reviews was not always straightforward. Some reviews had mixed tones, sarcasm, or ambiguous language, leading to misclassification.

Solution:

Advanced NLP techniques were integrated, such as lemmatization and using pretrained libraries like TextBlob and VADER, which handle context better. Additionally, human-in-the-loop validation was performed on random samples to refine labeling logic. This improved both the model’s accuracy and the intern’s understanding of natural language subtleties in real-world applications.

CHAPTER 7

HARDWARE AND SOFTWARE REQUIREMENTS

1. Hardware Requirements

A computer system with moderate to high specifications was required to ensure seamless operation across all tasks. The key hardware requirements included:

- Processor: Intel i5 or above / AMD Ryzen (multi-core) to support parallel processing tasks.
- RAM: Minimum 8 GB; 16 GB or more recommended for handling large datasets and machine learning.
- Storage: 256 GB SSD for faster performance; up to 500 GB total for large files and installations.
- Graphics: Optional GPU for advanced ML tasks or future deep learning projects.

2. Software Requirements

A wide variety of software tools and libraries were used throughout the internship, each serving specific purposes related to data analysis, machine learning, visualization, and natural language processing.

- Operating System: Windows 10/11, macOS, or Linux (Ubuntu preferred).
- Python & Libraries: Python 3.8+ with libraries such as pandas, numpy, scikit-learn, matplotlib, seaborn, nltk, wordcloud, and pyspark.
- IDE & Tools: Jupyter Notebook, Google Colab, VS Code, or PyCharm for scripting and development.
- Apache Spark: Version 3.x with PySpark, configured via SparkSession (Java 8 or 11 required).
- ML & NLP Tools: scikit-learn for ML models; NLTK and TextBlob for sentiment analysis.
- Visualization: Tableau Public or Power BI Desktop for dashboard creation.
- Other: A stable internet connection and modern browser for installations, cloud tools, and data access.

CHAPTER 8
CONCLUSION

This internship provided a comprehensive, real-world learning experience in the fields of big data, machine learning, data visualization, and natural language processing. Working on Amazon Prime

Video data across four structured tasks allowed me to develop end-to-end data science skills, from data ingestion and cleaning to model building, dashboarding, and sentiment analysis.

In Task 1: Big Data Analysis, I learned how to efficiently process large-scale datasets using PySpark, which helped me understand distributed computing and parallel processing—critical skills for enterprise-level data handling.

Task 2: Predictive Analysis focused on building machine learning models for classification. It strengthened my understanding of feature engineering, model evaluation, and the predictive power of structured data. I applied logistic regression and gained insights into model performance through metrics like accuracy and confusion matrices.

Task 3: Dashboard Development enhanced my ability to communicate insights visually. Using tools like Tableau or Power BI, I designed interactive dashboards that translated raw data into actionable business intelligence for non-technical stakeholders.

Task 4: Sentiment Analysis introduced me to working with unstructured data. By applying NLP techniques such as tokenization, lemmatization, and TF-IDF, I successfully classified user sentiments from text reviews, offering valuable insights into customer perceptions.

Throughout the internship, I improved both technical skills—Python, PySpark, ML, NLP, and BI tools—and soft skills like analytical thinking, problem-solving, and storytelling with data. Overcoming challenges such as data quality issues and model tuning gave me confidence to handle complex, real-world scenarios.

In conclusion, this internship was not just a technical project, but a transformative journey that bridged theory with practice. It has equipped me with a solid foundation in data science and inspired me to take on more impactful roles in the future